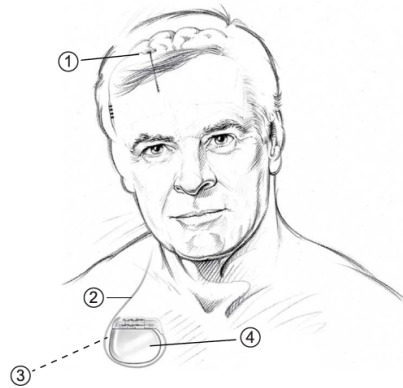


## ADVANCED MRI FOR DBS: UCSF INTERNAL RESEARCH PROTOCOL

### 3T BRAIN MRI with DBS implant



**Figure 1.** Medtronic DBS System components.

- |                            |                                                         |
|----------------------------|---------------------------------------------------------|
| ① Lead and burr hole cover | ③ Pocket adaptor (not present in all implanted systems) |
| ② Extension                | ④ Neurostimulator                                       |

**STUDY GOAL:** To study the underlying mechanisms of implantable deep brain stimulation (DBS), by evaluating fMRI data collected while therapy is ON, OFF, or in cycling mode.

**TARGET POPULATION:** Parkinson's Disease, Obsessive-Compulsive Disorder, Dystonia

**SCANNER:** GE 3T MR750 at UCSF Byer's Hall

**SAFETY STATEMENT:** The study will be conducted in accordance with FDA guidelines for imaging device B35200 (Percept PC) and B35300 (Percept RC). See [www.medtronic.com/mri](http://www.medtronic.com/mri) and enter device 'B35200' or 'B35300' to access safety requirements. Any off-label scanning will be approved by the MRI safety committee.

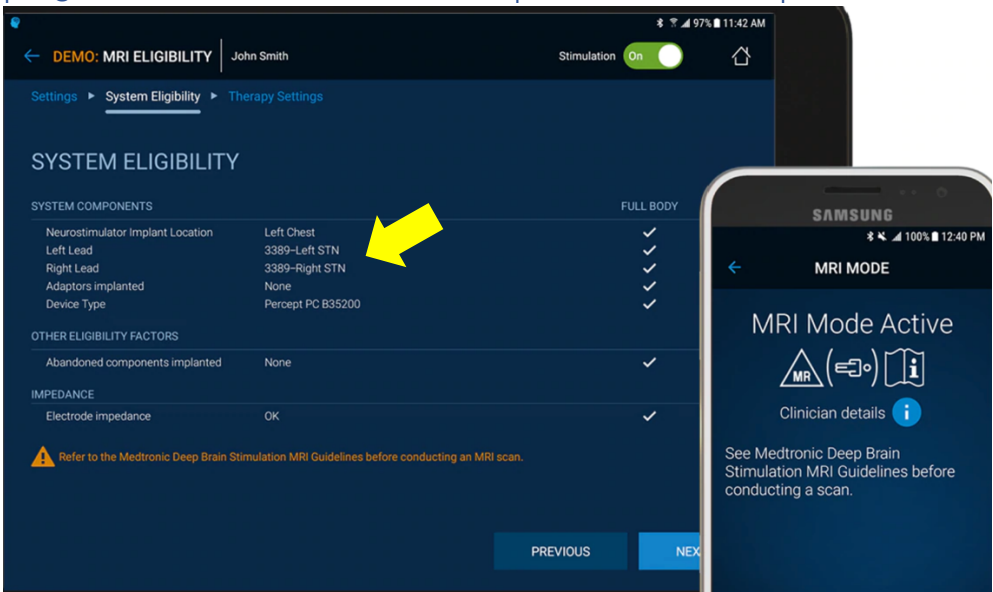
**SAFETY PERSONNEL:** PI, Neurologist or Psychiatrist (versed in DBS and disorder), Radiologist or Physicist (versed in MR safety), MRI technician, Nurse

### PRIOR TO ARRIVAL:

| Step                                                                                                                                | Done? |
|-------------------------------------------------------------------------------------------------------------------------------------|-------|
| pre-screen patient with MRI screening form                                                                                          |       |
| fill DBS MRI eligibility form for signing by physician; submit to MRI facility*                                                     |       |
| request scanner time; pend apex order for signing; update wiki                                                                      |       |
| patient meets with clinician for clinical assessment; <u>bipolar</u> therapy setting evaluated during visit will be used during MRI |       |
| ask patient to arrive with patient programmer, patient ID card, and implant devices charged (if applicable)                         |       |

\*confirms full-body system eligibility for 3T scanner. See [www.medtronic.com/mri](http://www.medtronic.com/mri) and enter device 'B35200' or 'B35300' to access form.

**IN CONSENT ROOM (~30 mins):**

| Step                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Done? |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| have patient fill and sign MRI screening form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| review exam details and risks with patient; consent/HIPPA if not already                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |
| confirm current medications and time since last dose                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| <p><u>clinician</u> review MRI eligibility report with patient:</p> <ol style="list-style-type: none"> <li>confirm patients name</li> <li>review report date; since report date, ask patient if they've had a fall, physical trauma, DBS revision surgery, or major changes to settings</li> <li>confirm neurostimulator(s) listed on report are correct by reviewing the programmer screen, serial number, patient ID card, implant location</li> </ol>                                                                                                                                                                                                                                                      |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |
| <p>d. confirm <b>bipolar</b> therapy settings to be used (by group name or letter)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |
| <p>check for short or open circuit:</p> <ul style="list-style-type: none"> <li>measure monopolar impedance between each electrode and neurostimulator, and bipolar impedance between all electrode pairs</li> <li><b>B33005, B33015 leads (SenSight models):</b> <ul style="list-style-type: none"> <li>&gt;8000 ohms or &lt;350 ohms = possible open/short circuit for monopolar impedance</li> <li>&gt;10000 ohms or &lt;350 ohms = possible open/short circuit for bipolar impedance</li> </ul> </li> <li><b>3387, 3389, 3391 leads (Activa models):</b> <ul style="list-style-type: none"> <li>&gt;2000 ohms or &lt;250 ohms = possible open/short circuit for monopolar impedance</li> </ul> </li> </ul> |       |

|                                                                                                                                                                                                                                                                                              |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <ul style="list-style-type: none"> <li>o &gt;4000 ohms or &lt;250 ohms = possible open/short circuit for bipolar impedance</li> <li>▪ if open or short circuit verified and cannot be troubleshoot based on information code and Medtronic Technical Services, <u>do not</u> scan</li> </ul> |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**PRIOR TO GOING INTO SCAN ROOM (~30 mins):**

| Step                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Done? |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| have patient change and remove metal objects; keep patient programmer (smartphone) accessible but <b>do not</b> bring in scanner                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| take patient temperature; <b>do not</b> perform MRI if body temperature > 100 °F                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| remind patient that they: <ul style="list-style-type: none"> <li>▪ <u>can hear and speak to us</u> in between the loud sounds, and <u>notify us to stop the scan</u> during the loud sounds by squeezing the ball</li> <li>▪ <u>may feel heating</u> at the neurostimulator site during the MRI. If it causes discomfort → squeeze ball → scan will be stopped</li> <li>▪ <u>may feel tugging and/or vibration</u> of the neurostimulator. If it causes considerable discomfort → squeeze ball → scan will be stopped</li> </ul> |       |
| set device to <b>MRI mode</b> and first <b>bipolar</b> setting for scan; if therapy will be off or cycling on/off remind patient that their symptoms may return abruptly<br><b>Note: if cycling DBS during fMRI, have a timer in hand. Start cycling mode and timer at same time. Cycling starts with DBS ON first. Prescan fMRI sequence so it can start on even or odd minute of DBS cycling. Mark the minute it starts on (and the preset duration of each cycle e.g., 30s) for GLM analysis.</b>                             |       |
| have neurologist (or psychiatrist) quickly assess patient symptoms                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |

**DURING SCANNING (30 mins of active scan time):**

**3T scanner requirements for Medtronic Percept PC/RC (B35200/B35300)**

*Note: there are other scanning options in the Medtronic manual.*

- MRI scanner manufacturer: any
- MRI system type: 3T horizontal cylindrical system; H-imaging; 128 MHz
- Maximum spatial field gradient: 20 T/m (2000 gauss/cm)
- RF coils: whole body transmit coil (integrated transmit coil) with any receive coil
- RF power: B1+rms ≤ 2.0 μT (**SenSight leads**) or ≤ 2.5 μT (**Activa leads**); SAR ≤ 1.0 W/kg
- Operating mode: First Level Controlled mode
- Maximum gradient slew rate: 200 T/m/s or less per axis
- Active scan time limits: do not exceed 30 minutes during a 90-minute window

**Note:** 3T systems may using two transmit channels (or fewer) may operate in Multichannel-2 or CP configurations. Systems with >2 transmit channels have not been studied, but such systems could be operated in CP or MC-2 configurations if available.

| Step                                                                                                                                                                                                                                             | Done? |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| have patient lie in scanner in supine position; <b>do not</b> use blankets for warmth; do not provide audio or visuals (unless task-based fMRI is being performed. Task will be explained to patient and equipment will be set up prior to scan) |       |
| load in patient information for scan via accession number                                                                                                                                                                                        |       |
| load protocol; confirm total scan time is 30 minutes or less                                                                                                                                                                                     |       |
| <b>set First Level Controlled operating mode</b>                                                                                                                                                                                                 |       |
| run localizer* and review artifacts of DBS components; check in with patient                                                                                                                                                                     |       |
| run T1* considering artifacts when selecting FOV; check in with patient                                                                                                                                                                          |       |
| run fMRI* copy FOV from T1; check in with patient                                                                                                                                                                                                |       |
| remove patient from scanner room by detaching bed                                                                                                                                                                                                |       |
| have clinician program next <b>bipolar</b> therapy setting; if therapy will be off or in cycling mode remind patient that their symptoms may return abruptly.                                                                                    |       |
| reattach patient to scanner and run next fMRI; check in with patient; <b>do not bring patient programmer (smartphone) in scanner room</b>                                                                                                        |       |
| remove patient from scanner and repeat until exam is complete                                                                                                                                                                                    |       |

\* Monitor SAR window throughout

\* Specific sequence parameters:

- T1: typical parameters, 1x1x1mm resolution (4 min)
- fMRI T2\*w EPI: 2.5x2.5x3mm resolution, 2500/20ms TR/TE, 150 time points (5 min x N runs)

## **FOLLOWING SCANNING:**

| Step                                                                                          | Done? |
|-----------------------------------------------------------------------------------------------|-------|
| have clinician reprogram patient's original therapeutic setting and reassess for potential AE |       |
| review the final T2*-weighted fMRI images for potential AE                                    |       |
| push T1 and last fMRI scan to PACs for read → push all data to RESEARCH                       |       |
| Coordinator calls patient one week after scan to check in                                     |       |